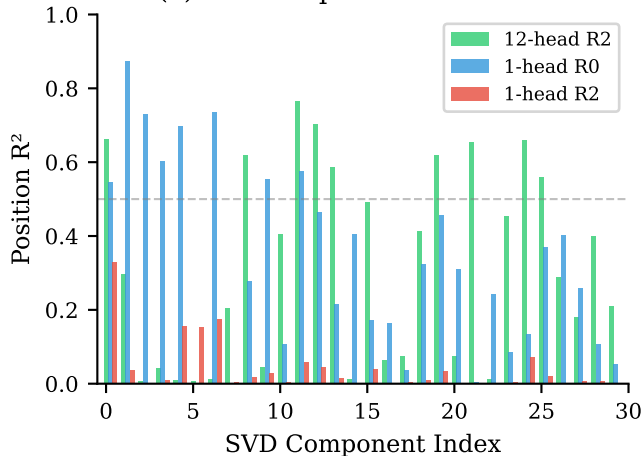
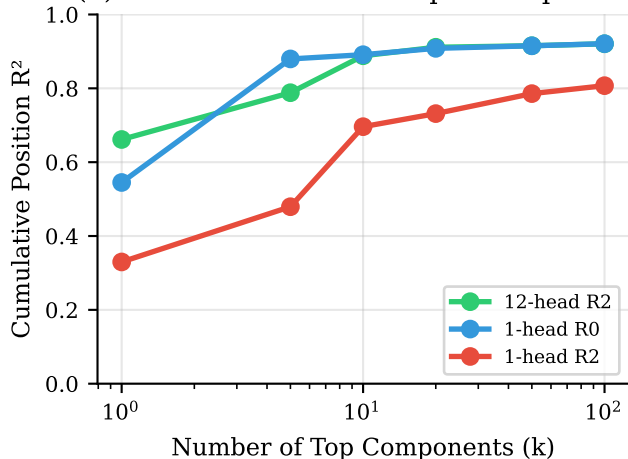
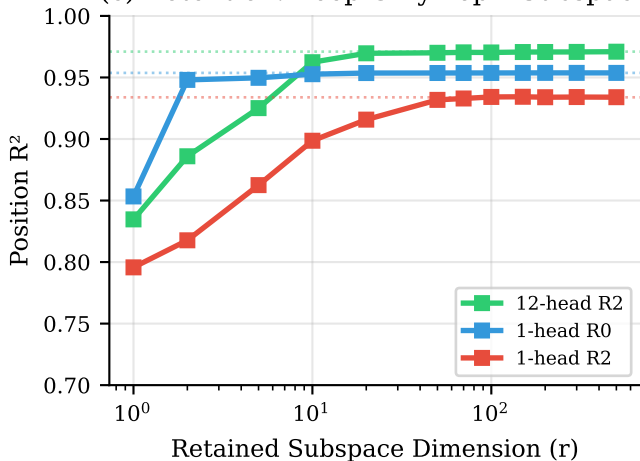


(a) Per-Component Position  $R^2$ (b) Cumulative  $R^2$  from Top-k Components

(c) Retention: Keep Only Top-r Subspace



(d) Ablation: Remove Top-r Subspace

